

SMART CLASSROOM AUTOMATION SYSTEM Project Report

1. Introduction The Smart Classroom Automation System enhances classroom security, attendance management, and energy efficiency using biometric authentication and motion-based automation.

2. Objectives - Automate attendance using fingerprint sensor.

- Reduce electricity wastage using motion-based appliance control.
- Enhance classroom security.
- Provide a scalable, low-cost automation solution.

3. System Overview The system integrates a fingerprint sensor, PIR/IR motion sensors, relay modules, and a microcontroller (Arduino/ESP32). It marks attendance and controls lights/fans automatically.

4. Hardware Requirements - Arduino Uno / ESP32

- Fingerprint Sensor (R305 / AS608)
- PIR/IR Motion Sensors
- Relay Modules
- 12V Fans & Lights
- Power Supply (5V & 12V)

5. Working Principle 5.1 Attendance System User places a finger on the sensor → fingerprint is verified → attendance recorded.

5.2 Motion-Based Automation Motion detected → relay activates lights/fans → no motion → appliances turn OFF.

6. Applications - Schools & colleges

- Smart labs
- Training centers

7. Conclusion The Smart Classroom Automation System is a reliable, scalable, and energy-efficient solution for modern educational environments.